

SHOGUN design pack (2026-10-06)

Working name SHOGUN. A fan instrument in the Jidai Collection. Design pack, voice engine, a GraphForge pin for the clock switch, and a placeholder plate.

SHOGUN is a drum computer and two note voices, an original Martial Systems design inspired by classic analog drum machines. The jobs of the voices follow the design pack (its sources are cited in `METHODOLOGY.md`) and the design brief. The panel, when it is drawn, is a new face. This tree has the design pack, the voice engine in `engine/`, the clock-switch pin in `forge/`, and a placeholder VST3 under `plugin/`. Panel art is a later commit. The editor drawn now is a flat plate so the instrument can be loaded in FL Studio.

Revisions

- 2026-10-06: first pack. Sources, clock bypass, voice equations, sequencer, tests, and the PDF of the same text.
- 2026-10-06: voice engine. Framework-free C++ in `engine/`, INT and EXT trigger paths, and the named tests in `tests/voices.cpp`.
- 2026-10-06: clock switch pinned under `forge/`. A Trig cable cannot force EXT. Printed 48 kHz rows stay named tests.
- 2026-10-06: placeholder VST3. Flat plate, knobs, levels, a 16-step grid, and TRIG buttons. `make plugin` builds a universal instrument. Panel art stays later.
- 2026-10-07: engine fixes and the web panel. Clock, conga, held-note and tie fixes listed in `BUGS.md`, with tests. `make web` compiles the engine to WebAssembly, checks it against the native build, and writes `web/shogun.html`.
- 2026-10-07: quiet drum voices end at 1e-6. BD1 Dist 0 is a bypass and drive is 9u above it. Main-only centre voices are equal power. Shuffle follows the clock step. Cymbal noise is a fixed mix. The pack states the rest rules, EXT lead and bass, and the CLK IN, RST IN, and RUN IN jacks.
- 2026-10-07: BD1 noise rides the body envelope, so a noisy kick ends with its body. The pan law is equal power: every centre, toms and clap tail included, is 0.707107 on each side.
- 2026-10-07: one decay law for every drum, $\tau = 8 \text{ ms} * \exp(4.5 \text{ u})$, and a voice is 0 under 1e-3. BD2 at Decay 127 holds; a tom at Decay 127 rings 4 s and ends. The snare bend is its own drop to Tune with an 80 ms floor.
- 2026-10-07: track solo. The soloed voice alone reaches its pair and the main; the others keep running at 0. Mute wins.
- 2026-10-07: a new engine loads the init kit and pattern. STAND-IN knob values set by ear. `reset()` still clears.
- 2026-10-07: factory patterns and kits cleared for now. The engine, plugin and web page start on INIT, an empty bar, with the INIT kit. New factory content will be written later.
- 2026-10-07: density pass. Shaped bodies ($\tanh(k \sin)$) on BD1, BD2, SD, and the toms, BD2 and tom slow FM, a 1 ms BD1 click, SD noise through a ducked 4-pole, six-square metal stacks with a band-pass on the hats and cymbal, and clap bursts fixed at 3 ms with a delayed filtered tail. STAND-IN values set by ear.
- 2026-10-07: Wave on BD1, BD2, and the three toms: a six-cell wave folder on the body oscillator, CC 0 bypass, default CC 32, g 0.5 to 4. Dist stays after Wave. Hats, clap, cymbal, and maracas have no Wave.

What this is

A stand-in design. Every coefficient in `SCHEMATICS.md` is marked STAND-IN. Where the English manual, the German manual, and the control-change list disagree, both wordings are quoted and the pick is marked ASSUMED or CHOICE. No service schematic was in the sources. The equations are functional blocks, not a transistor circuit.

The instrument this pack describes:

- 14 drum tracks and 2 note tracks
- An internal sequencer with patterns of 1 to 32 steps

- A clock switch, INT or EXT, that connects or disconnects that sequencer from the voice triggers. The switch chooses. INT ignores Trig jacks. EXT ignores the pattern. A plugged Trig cable does not force EXT.
- Paired outputs plus a main mix

Files

File	Role
README.md	This front page
METHODOLOGY.md	Sources, quotations, and the choices
SCHEMATICS.md	Block, sample equation, knob range, and a 48 kHz example per voice
BUILD_GUIDE.md	Engine layout, PDF build, and the panel rule
TESTPLAN.md	Named tests and the numbers they lock
REPO_SETUP.md	Tree, git, and the collection link
engine/	Voice engine, clock, pattern, and outs. No JUCE
tests/voices.cpp	The named checks, compiled by the Makefile
plugin/	Placeholder VST3. JUCE wraps the engine. The plate is stock controls
Makefile	<code>make test</code> builds the engine checks and runs the clock-switch law. <code>make plugin</code> builds the VST3
forge/	GraphForge pin for the clock switch. Printed sample rows stay in <code>tests/voices.cpp</code>
docs/SHOGUN_Design_Pack.pdf	The same documents in one PDF
BUGS.md	Engine bugs fixed, engine notes, and open questions on the pack
web/	The playable panel. <code>web/wasm/</code> wraps the engine for the browser, <code>web/page/</code> is the panel source, <code>web/shogun.html</code> is the built page

Rebuild the PDF with:

```
python3 scripts/build_design_pack_pdf.py
```

The Jidai Collection

BUSHIDO and RONIN share one patch format and one cable feel. JIDAI RACK compiles those two engines. SHOGUN's engine is in this repository. It is not compiled into that rack.

- **BUSHIDO**: a 3 x 12 analog step sequencer with real patch cables. <https://github.com/martialsystems/bushido>
- **RONIN**: a semi-modular synthesizer you patch as an effect. <https://github.com/martialsystems/Ronin>
- **JIDAI RACK**: BUSHIDO and RONIN in one rack, in one plugin. <https://github.com/martialsystems/jidai-collection>

Legal

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SHOGUN is an original Martial Systems design inspired by classic analog drum machines. Martial Systems is not affiliated with or endorsed by any drum machine maker. The design pack quotes short phrases from published manuals for the parameter disputes and cites them in METHODOLOGY.md. It does not reproduce the manuals, and it does not use any maker's logo or marks.

Methodology (2026-10-06)

Status

This is a stand-in design for a fan instrument. The voices are analog blocks: oscillators, envelopes, noise, and a clip. Hats and cymbals are square or metallic stacks plus noise. No sample is in the voice model. No coefficient is a measured part value. Each one is marked STAND-IN in SCHEMATICS.md.

No service schematic was found in public, and none is invented here. A block diagram is the farthest the circuit claim goes.

Sources

Read on 2026-10-06.

1. German primary. *MFB Tanzbär Bedienungsanleitung*, 17 PDF pages, two printed pages on most sheets, printed page numbers through 31. PDF metadata: CreationDate D:20131024121806+02'00, Creator Adobe InDesign CS5 (7.0), Producer Adobe PDF Library 9.9. Text extracted from the file published at https://media.djmania.net/manuales/pdf/Manual_MFB_Tanzbar.pdf. The MIDI appendix is printed page 31 of this file. That appendix is the control-change list used below.

2. English primary. *MFB Tanzbär User Manual*, the 17-page English edition with the same contents list (overview through MIDI implementation). Printed pages quoted below were read from <https://www.manualshelf.com/manual/mfb/tanzb-ar/user-manual-english/page-3.html> and the neighbouring pages of that edition (pages 4, 6, 8, 9, 11). A separate English PDF file was not downloaded. The contents list and the instrument chapter match the German edition's structure.

3. Published control-change republication. <https://midi.guide/d/mfb/tanzbar-1/>, last update 12 September 2024, 64 parameters. This is the list identified by BD1 Attack on CC 2, Decay on 64, Pitch on 65, Tune on 3, Noise on 4, Filter on 5, Dist on 6, and Trigger on 66. Where it disagrees with the 2013 appendix, the appendix wins and the row is marked.

A secondary web conversion at <https://manuals.plus/mfb/mfb-tanzbar-analog-drum-machine-manual> was read only for the Data-versus-Sound conflict named in the brief. It is not a primary.

Tanzbär 2, Tanzmaus, and Tanzbär Lite were not used.

Marks

- STAND-IN: a number chosen so the equation is definite. It is not a capture.
- CHOICE: the sources, or the sources plus the brief, support this reading, and the pack uses it.
- ASSUMED: the sources disagree, or they are silent, and a single reading is still required.

Parameter names

Both primary instrument chapters agree on bass drum 1.

English, printed pages 14 to 15: Attack is the level of the attack transients. Decay is volume decay time. Pitch is time and modulation intensity of the pitch envelope. Tune is pitch. Noise is noise level. Filter is the sound of the noise signal. Data is distortion level (printed "Distortion"). Sound selects 1 of 16 different attack transients.

German, printed pages 14 to 15: Data "regelt die Intensität des Verzerrers". Sound: "16 verschiedene Attack-Transienten stehen zur Auswahl". Filter "beeinflusst den Klang des Rauschanteils".

The 2013 appendix names the same jobs as BD1_DIST on CC 6 and BD1_TRIGGER on CC 66. It has no row named Sound.

The secondary conversion says BD1 Data is "Sound of noise signal". That sentence swaps Data with the job both primaries give to Filter.

CHOICE: Sound, which the appendix calls BD1_TRIGGER, selects the transient, index 0 to 15. Data, BD1_DIST, is the clip. Filter stays on the noise.

Quoted disputes

Individual outputs, overview against the jack list

English overview, printed page 4: "Individual outs (in pairs except claps)."

German overview, printed page 4: "Einzelausgänge (bis auf Bassdrum und Clap paarweise ausgeführt)."

Both rear-panel lists then say: BD left is bass drum 1 and BD right is bass drum 2. SD/RS is snare and rim. HH/CY left is open and closed hat, right is cymbal. The clap out is stereo, with the attack transients spread in the field. TO/CO spreads the three toms or congas. CB/CL left is clave, right is cowbell.

The two overview sentences disagree with each other. The jack lists agree with each other and with the brief.

CHOICE: follow the jack list. Maracas, the lead voice, and the bass voice are not on that list. They go to the main mix only.

Cable in an individual out

English printed page 7: "If you plug in a cable on an instrument out, the sound is erased from the main out."

German printed page 7: "Wird ein Kabel in einen Einzelausgang gesteckt, werden die betreffenden Sounds im Hauptausgang gelöscht."

The two primaries agree: the hardware removes those sounds from the main output.

CHOICE for SHOGUN: an individual out is a tap. The voice stays in the main mix. The hardware sentence is recorded so the difference is explicit. TESTPLAN.md locks it as `testIndividualOutStaysInMix`.

Seventeen names, fourteen tracks

Both overviews say 17 drum instruments and 14 tracks that trigger drum instruments. The instrument chapter lists 14 drum names: BD1, BD2, SD, RS, CY, OH, HH, CL, CP, LTC, MTC, HTC, CB, MA. The two note tracks are extra.

ASSUMED: each tom track has a tom mode and a conga mode, and the overview counts those as six names on three tracks. 11 other drums plus 6 is 17 names on 14 tracks. The manuals do not write that sum.

Tom and conga buttons

English, printed pages 15 to 16, gives LTC, MTC, and HTC the same buttons: step button 12 toggles tom and conga, step button 13 enables noise.

German, printed pages 15 to 16: LTC uses the lit LEDs 12 and 13, MTC uses 13 and 14, HTC uses 14 and 15.

The appendix gives each voice its own noise switch and its own tom/conga switch, plus one shared TOM_NOISE on CC 84.

ASSUMED: the voice model uses the per-voice switches and the shared level. The button numbers are a panel procedure, and the two editions do not agree, so they are not part of the equation.

Track length

German overview: "Länge der Spur (1 bis 32 Steps)."

German record chapter, printed page 23: each track may have an individual length "zwischen einem und 16 Steps".

English overview: step number 1 to 32, with four scale settings.

CHOICE: pattern length and per-track length are integers from 1 to 32, which is the overview and the brief. Steps 1 to 16 are the A half. Steps 17 to 32 are the B half. The "1 to 16" sentence matches the sixteen step buttons on the A half.

Scale and length

Both manuals tie scale to a default pattern length: 32nd notes give 32 steps, 16th-note triplets give 24, 16ths give 16, 8th-note triplets give 12. Measure can then set another count. The bar still takes the same time, so a 32nd scale runs the sequencer twice as fast as a 16th scale.

The brief asks for scale as step length, and for patterns of 1 to 32 steps as their own setting.

CHOICE: in SHOGUN, scale sets only how many steps fit in a quarter note. Pattern length is a separate integer from 1 to 32. The hardware coupling is not copied. This split is ASSUMED relative to a machine whose scale button also picked the length.

Cowbell resolution

Both instrument chapters: Data offers 16 tunings, and Sound is the decay time.

Appendix: CB_Tune on CC 85, value 0 to 127, and CB_Decay on CC 86, value 0 to 127.

ASSUMED: Tune and Decay are continuous 0 to 127, under the appendix names. A 16-step control is a quantizer of Tune, step i mapping to $CC \text{ round}(i \times 127 / 15)$. The brief's two detuned squares are the oscillator. The manuals do not name the waveform.

Clap type and count

Both instrument chapters: Decay is the tail, Filter is colour, Attack is the level of the attack transients, Data is the number of those transients, Sound is 16 different transients.

The record chapter's hidden-parameter paragraph names BD1, the toms and congas, and the cowbell. It does not name the clap. English printed page 20 and German printed page 20 both still say flam cannot be programmed on Clap, CV1, or CV2/3.

Appendix: CP_DECAY 75, CP_FILTER 18, CP_ATTACK 76, CP_TRIGGER 77. No row is named count, and no row is named Sound.

ASSUMED: CP_TRIGGER is the type, on the same pattern as BD1_TRIGGER. The count is Data, an integer from 1 to 8, and it has no published CC. The record paragraph's shorter instrument list is treated as incomplete next to the instrument chapter.

Set select and one track-delay label

2013 appendix: Set Select, controller 0, value 0 to 2.

midi.guide: Set Select has no controller number, and the range column says 0 to 127.

2013 appendix: Track Delay HH, controller 97.

midi.guide labels controller 97 "Track Track HH". The number matches.

CHOICE: controller 0 with values 0, 1, and 2, and the name Track Delay HH for 97.

Clock bypass

INT and EXT are SHOGUN. They are not a switch in the Tanzbär manuals. Those manuals have MIDI clock, an analog sync jack that can be an input or an output, and a manual-trigger mode. None of those is this switch.

The switch is the only control that selects the master.

- INT: the internal sequencer triggers the voices. Its tempo is the host tempo while the host is playing, otherwise its own tempo. A cable in a Trig jack does not fire the voice.
- EXT: the internal sequencer does not trigger the voices. Step on and off, flam, and accent from the pattern are ignored for triggering. Knob values still apply. Each voice fires from its own Trig jack. Accent may arrive as velocity on that jack.
- EXT does not stop the clock. The step counter and the clock display keep advancing. Switching back to INT connects the triggers to the step the clock has reached.
- A cable in a Trig jack does not force EXT.

ASSUMED: "resumes" means the counter never froze. The alternative, a frozen counter that continues from the step where EXT was engaged, would stop the clock the brief says keeps running.

Voices the manuals do name

The instrument chapter gives the jobs below. Ranges and waveforms that the chapter does not give are in SCHEMATICS.md, marked there.

- BD1: attack level, decay, pitch envelope time and depth on one knob, tune, noise level, noise filter, distortion, and 1 of 16 attack transients.
- BD2: decay up to a steady tone, tune, and tone as the attack level. The chapter does not give BD2 a noise, a filter, a distortion, or 16 transient types. It is a second kick.
- SD: tune shared by two tones, detune of tone 2, snappy noise level, noise decay, blend of the two tones, decay of both tones, and pitch-envelope intensity.
- RS: pitch only. The brief asks for a short pitched click.
- CY: decay, blend of both signals, and pitch or colour.
- OH and HH: separate decays, and one pitch or colour shared by both. The appendix exposes that colour as HH_TUNE on CC 73. There is no OH tune row.
- CL: tune and decay.
- CP: tail decay, filter, attack level, a count of attack transients, and 16 transient types. The clap out spreads those transients. The brief adds the spacing, about 10 to 12 ms, which the manuals do not print.
- LTC, MTC, HTC: tune, decay up to a steady tone, noise on or off, tom or conga, and one noise level for all three. Bend is available. The tom out spreads the three.
- CB: tune and decay in the appendix. The brief specifies two detuned squares.
- MA: decay only. The brief specifies filtered noise.
- Lead and bass: the overview says one parameter each. Bass Data is "Filter cutoff or CV 3 value" in the English chapter. Setup plays a steady 440 Hz and tunes both voices with Data. The waveform is not named. ASSUMED waveform: a band-limited saw for each. ASSUMED lead parameter: a one-pole tone, because the chapter names one parameter and does not name the lead knob.

Closed hat kills open hat. The pages read do not state a choke time. The brief requires the choke, so the pack defines it: the closed-hat trigger clears the open-hat envelope on that sample.

Sequencer the manuals do name

- Patterns live in 3 sets of 3 banks of 16, which is 144. That memory map is the hardware's. SHOGUN stores patterns without a 144-slot cap in this pack. The set-select controller is still defined.
- A and B. A pattern longer than 16 steps has a B half, and the machine walks it. A/B can also alternate. Copy of A onto B exists on the hardware. Chain is runtime only: the English overview calls it a fill that is not storable, and the German overview says the chain is "nicht speicherbar".
- Per track: length, shuffle, shift, mute. Shuffle in record mode is per track, 16 intensities. Shuffle in play mode is global and does not replace the stored per-track values when it is released. Shift is controllers 89 to 104, described as a small delay of the whole track. No millisecond range is printed.
- Per step, drums: on or off, one of three accent levels (off or soft, green or medium, red or loud), flam on 16 patterns, bend where the instrument has it, and an extra sound value where the instrument chapter has one. Bend instruments are BD1, BD2, SD, LTC, MTC, and HTC. Flam is unavailable on clap and on both note tracks.
- Per step, note tracks: on or off, pitch over three octaves, accent on the bass track, and a second value on the bass track that is CV3 and the internal bass cutoff. A/B enters a silent step. Select ties steps into a longer note.
- Level pots and the master volume are not stored in the pattern.

- Manual record mode shares one knob set across steps. Accents and flams can still differ per step. Step and jam modes can store different knob values per step.

Tempo

English play chapter: the tempo knob covers approximately 60 to 180 BPM, and there is no tempo readout.

CHOICE: SHOGUN's internal tempo uses those endpoints as exact bounds, default 120. The clock display is part of SHOGUN. The hardware has no readout.

Gate level

A Trig jack is a rising edge through 1.0 V on a gate that rests at 0 V and is high at 5 V. That 1 V threshold is the collection's existing gate rule between RONIN and BUSHIDO. Velocity, when the sender has one, is an integer 0 to 127 riding with the edge. A plain gate with no velocity uses 127.

ASSUMED: the plain-gate default of 127. The Tanzbär manuals say drum notes carry velocity. They do not define a jack that has no velocity byte.

Schematics (2026-10-06)

Functional blocks and sample equations. Coefficients are STAND-IN. Sample rate in every numeric example: 48,000 Hz. Arithmetic is IEEE-754 double. A later test may treat the printed decimals as exact within 1e-5.

Shared helpers:

```
u(cc) = cc / 127
fs = 48000
pi = 3.141592653589793
```

cc is an integer from 0 to 127. A per-step override, when the record mode stored one, replaces the live knob for that step. Manual record mode does not store per-step knobs.

A rising trigger at sample n0 resets that voice's phase to 0 and restarts its envelopes at n0. Until the first trigger, a cleared voice outputs 0.

Decay law. Every drum Decay knob, and the snare's Tone Decay and SN Decay, sets its time constant the same way. STAND-IN:

```
decay_tau(u) = 0.008 * exp(4.5 * u)      # seconds
```

u = 0 is 8 ms, u = 0.5 is 75.902 ms (CC 64 is 77.259 ms), and u = 1 is 720.137 ms. The clap tail uses it. The clap bursts keep their fixed 3 ms. RS has no Decay knob. BD2 at Decay 127 is the one exception: it holds. A tom at Decay 127 rings for 4 s and then ends.

A drum voice ends when every envelope it has is under 1e-3. From that sample, that sample included, it outputs 0 until its next trigger. Only a held BD2 does not end, because its envelope stays at the sustain.

Rests. A drum step that is off fires nothing, and a drum voice that is still ringing rings out. A note-track rest releases the note that is sounding, with the release in Lead and bass.

Velocity law

Pattern accent index a is 0, 1, or 2. English printed page 20: LED off is soft, green is medium, red is loud.

```
g_accent(0) = 0.55
g_accent(1) = 0.78
g_accent(2) = 1.00
```

STAND-IN.

EXT velocity v , an integer 0 to 127:

$$g_vel(v) = 0.15 + 0.85 * (v / 127)$$

STAND-IN. $v = 0$ gives 0.150000. $v = 100$ gives 0.819291. $v = 127$ gives 1.000000.

INT multiplies the voice by g_accent and ignores Trig velocity. EXT multiplies by g_vel and ignores the pattern accent. The gain multiplies the voice output after the voice equation.

Clock

Steps per quarter note, from the four scale settings in both manuals:

Scale	Steps per quarter	Samples per step at 120 BPM
32nd	8	3,000
16th triplet	6	4,000
16th	4	6,000
8th triplet	3	8,000

$$period = fs * 60 / (bpm * steps_per_quarter)$$

At 120 BPM and a 16th scale, period is 6,000 samples. Internal bpm is clamped to 60 through 180. Default 120.

The step counter increments on each period boundary in INT and in EXT. The clock display shows $(counter \bmod pattern_length) + 1$. $pattern_length$ is the bar, an integer from 1 to 32.

Each track stores its own length L , also 1 to 32. On global counter c the track reads step $c \bmod L$. A shorter track cycles against the bar. A longer track has steps past one bar, which is how a track reaches the B half when the bar is shorter. The track fires when that stored step is on. It does not also have to fall inside the bar length.

Shuffle intensity s is an integer 0 to 15. STAND-IN. The delay goes on odd clock steps, counter c odd, whatever the track's length. A track of odd length still swings on the beat, so its own step 0 is delayed on every other pass:

$$delay_samples = (s / 15) * (period / 3)$$

$s = 0$ is no delay. $s = 8$ at period 6,000 delays an odd clock step by 1,066.666667 samples. The fractional sample is kept and the trigger fires on the sample that contains the fractional boundary. Play-mode global shuffle, while held, uses one s for every track and does not write the pattern.

Track shift, controllers 89 to 104, one per track including the two note tracks. STAND-IN:

$$shift_samples = round(u(cc) * 0.030 * fs)$$

Full scale is 1,440 samples, 30 ms. The whole track's triggers move later by that many samples.

INT and EXT

INT fires a drum voice when its step is on. The flam table and the accent index apply. The Trig jack is not read.

EXT does not fire from the pattern. Flam, step on and off, and pattern accent do nothing. The Trig jack's rising edge fires the voice, and g_vel supplies accent. Knobs, including per-step knob overrides if a later editor still wants them, remain in force. This pack's EXT path uses the live knobs, not the pattern's per-step sound overrides, because the brief says the pattern is ignored for step on and off, flam, and accent, and that knob values still apply. ASSUMED: "knob values" means the live knobs, not values stored on ignored steps.

Lead and bass in EXT follow the live gate on their Trig jacks. A rising edge opens the note at the live pitch with g_vel . The falling edge starts the release. The pattern's notes, rests, and ties do nothing in EXT.

Bend stored on a step is part of the pattern. EXT ignores it. The Pitch knob on BD1, BD2, SD, and the toms still applies, because that knob is a sound control, not the step bend.

Clock jacks

CLK IN, RST IN, and RUN IN are gate inputs at the same 1 V threshold as the Trig jacks. A rising edge on CLK IN advances the counter one step, and while CLK IN is patched the internal period does not. A rising edge on RST IN sets the counter to 0, so the next step is step 1. A rising edge on RUN IN starts a stopped clock or stops a running one. None of them changes the INT and EXT switch. Only the switch chooses.

Noise generator

One generator for the whole instrument, so examples reproduce.

```
state = (1664525 * state + 1013904223) mod 2^32
x = state / 2^32 * 2 - 1
```

Initial state is 1. The first draw yields $x = -0.527089$. Each noise consumer takes the next draw when it needs a sample. The maracas example below is the first draw of a cleared generator.

One-pole lowpass, state y previous 0 at a trigger:

```
a = exp(-2 * pi * fc / fs)
y[n] = (1 - a) * x[n] + a * y[n-1]
```

Output sum

Instrument level and master are linear gains, not stored in the pattern. Default level 1, default master 1. STAND-IN, they are plain multipliers.

Equal-power pan, p from -1 to 1:

```
gL = cos(pi / 4 * (1 + p))
gR = sin(pi / 4 * (1 + p))
```

STAND-IN. Every centred voice, $p = 0$, is 0.707107 on each side, which is $\sqrt{0.5}$, so it sits level with a hard-panned voice. $p = -0.7$ is $gL = 0.972370$, $gR = 0.233445$.

Voice	Pair	p
BD1	BD L	hard left, written only to L at gain 1
BD2	BD R	hard right, written only to R at gain 1
SD	SD/RS L	hard left
RS	SD/RS R	hard right
OH + HH	HH/CY L	summed, hard left
CY	HH/CY R	hard right
CP bursts	CP stereo	pan of burst i , below
CP tail	CP stereo	$p = 0$
LTC	TO/CO	$p = -0.7$
MTC	TO/CO	$p = 0$
HTC	TO/CO	$p = 0.7$
CL	CB/CL L	hard left
CB	CB/CL R	hard right
MA, lead, bass	main only	$p = 0$

The voice sample is after the accent or velocity gain and after the instrument level. Hard left adds that sample to the left channel of the pair and of the main, and adds 0 to the right. Hard right is the swap. A panned voice uses gL and gR on both the pair and the main. The master multiplies the main only. Pair jacks are before the master.

Solo. One voice at a time can be soloed, or none. While a voice is soloed, only that voice reaches its pair and the main. Every other voice still triggers and runs its envelopes, and outputs 0, so when solo is turned off each one is heard where it would have been. A muted track stays silent when it is soloed: mute wins. Solo is not stored in the pattern, and it does not change

the INT and EXT switch.

CHOICE: the main mix is that sum whether or not a pair jack is patched. Maracas, lead, and bass have no pair.

Flam table

Flam exists on every drum track except clap. Sixteen patterns, index k from 0 to 15. The manuals name 16 patterns and do not print the timings. STAND-IN offsets, in samples, from the step:

```
hits = 2 + (k mod 4)
gap = 180 * (1 + floor(k / 4))
offsets = 0, gap, 2*gap, ... while the list length is hits
```

Index 0 is samples 0 and 180. Index 15 is 0, 720, 1,440, 2,160, and 2,880. Each offset fires the voice again. A new fire restarts that voice's envelopes. Offsets may cross into the next step.

Wave

BD1, BD2, LTC, MTC, and HTC each have a Wave knob. Hats, clap, cymbal, and maracas do not. Wave is the Serge middle-section folder: six identical cells in series, ported from martialsystems/serge_middle (the cell, its Lambert W solver, and its drive peak table), not a second waveshaper.

Each cell, for v not 0, with VT = 0.02585, Is = 2.52e-9, eta = 1.68, R = 33,000:

```
z = (Is * R) / (eta * VT) * exp(|v| / (eta * VT))
v_out = sign(v) * (|v| - 2 * eta * VT * W(z)) # v_out(0) = 0
C6(v) = C(C(C(C(C(C(v)))))))
```

W is the principal Lambert W, solved by Halley on $w + \ln(w) - \ln(z) = 0$ so the exponential is never formed. The residual is under 1e-12 through 6 V and at 48 V.

```
g(cc) = 0 at CC 0, the bypass
g(cc) = 0.5 + 3.5 * (cc - 1) / 126 CC 1 to 127, 0.5 to 4. STAND-IN taper
fold(x) = x if wave_cc = 0
fold(x) = (4.379272 * P(1) / P(g)) * C6(5 * g * x) otherwise
```

x is the body oscillator, sin(phase), about +/-1, scaled to +/-5 V times g before the cells. P(g) is the serge_middle peak of |C6(v)| for |v| <= 5 g, linear between its 53 knots, so a full-scale x peaks at 1 after the cells at every g and Wave does not change the peak level. Default Wave is CC 32 (0.25), g = 1.361111.

The fold acts on the oscillator, before the body envelope, so the decay law and the 1e-3 end are untouched. On BD1 the Dist clip comes after Wave. The 4x oversampler, the 10 Hz DC block, and the knob smoother of serge_middle are not in this pass. A folded sine is odd, so it has no DC.

Measured on a held body at velocity 1, the share of spectrum energy above 1.5 times the fundamental is under 0.005 at Wave 0, 0.388 at the default, and 0.033 at CC 127. The folds sit in the bottom 5.2 V of the cell's input (fold roots 0.73, 1.55, 2.43, 3.34, 4.27, 5.22 V). Past that each cell is close to an inverter, so at g = 4 most of the 20 V swing passes through and, after the peak is brought back to 1, the folds are a ripple near each zero crossing. The densest setting is the low end of the knob.

The init kit at the default Wave: one BD1 hit peaks at 0.963 against 0.964 at Wave 0, and its RMS over 0.5 s is 0.106 against 0.155 (3.3 dB lower).

BD1

```
trig --> pitch and bend envelopes --> sine --> Wave --+
--> click[s], 1 ms -----+--> sum --> tanh clip --> gain
--> noise burst on the body env --> one-pole --+
```

Density pass, STAND-IN coefficients set by ear, no hardware capture. The clip is bypassed at Dist 0.

Decay does not reach a steady tone. The body envelope falls to 0 for every knob value.

```
f_tune = 35 * (140/35) ^ u(tune) # 35 Hz to 140 Hz
depth = 18 * u(pitch) # semitones, 0 to 18
```

```

tau_p = 0.012 + 0.25 * u(pitch)          # seconds
tau_bend = max(tau_p, 0.08)              # the step bend's own time, 80 ms floor
tau_b = decay_tau(u(decay))              # 8 ms to 720 ms
s = min(15, floor(trigger_cc * 16 / 128))
f_tr = 160 * (1.35 ^ s)
fc = 200 * (8000/200) ^ u(filter)        # 200 Hz to 8,000 Hz
drive = 9 * u(dist)                      # 0 at Dist 0, which is the bypass

shape(ph, k) = tanh(k * sin(ph)) / tanh(k)

p_env[n] = exp(-n / (fs * tau_p))
b_env[n] = exp(-n / (fs * tau_bend))      # 0 for every n when bend_st = 0
f[n] = f_tune * 2 ^ ((depth * p_env[n] + bend_st * b_env[n]) / 12)
body[n] = fold(sin(phase[n])) * exp(-n / (fs * tau_b))    # Wave, above
click[n] = wave_s(n) * (1 - n / 48) ^ 2    for n < 48, else 0
burst[n] = x[n] * exp(-n / (fs * tau_b)) * exp(-n / (fs * 0.015))
noise = u(noise_cc) * lowpass(burst[n], fc)
pre[n] = body[n] + u(attack) * click[n] + noise
y[n] = pre[n]                            if dist_cc = 0
y[n] = tanh(drive * pre[n]) / tanh(drive)  if dist_cc > 0

```

The click is 1 ms, one of 16 shapes. $s \bmod 4$ picks the wave and s sets its rate f_{tr} : 0 is $\sin(2 \pi f_{tr} n / fs)$, 1 is $\tanh(6 \sin(2 \pi f_{tr} n / fs)) / \tanh(6)$, 2 is $\cos(2 \pi f_{tr} n / fs)$, a hard tick, and 3 is a saw, $2 * \text{frac}(f_{tr} * n / fs) - 1$. STAND-IN.

The noise is a short burst under the body envelope, then the one-pole. The burst is 0 wherever the body is, so the voice is silent once the body is under $1e-3$. There is no separate noise decay knob.

phase advances by $2 * \pi * f[n] / fs$ after the sample is taken. n is samples since the trigger. $bend_st$ is the step bend in semitones. Like the snare, it is its own drop to Tune: its time is the Pitch time, but never under 80 ms while a bend is set. INT step bend is $bend_st = 12 * (2 * u(bend_cc) - 1)$. CC 0 is -12 semitones, CC 127 is +12, and CC 64 is $12 * (128/127 - 1) = 0.094488$ semitones. STAND-IN. EXT forces $bend_st$ to 0. The printed BD1 example uses $bend_st = 0$, noise CC 0, dist CC 0, and the default Wave, CC 32, so $y = pre$. $drive = 9u$ starts at 0.070866 for CC 1, where $\tanh(drive * pre) / \tanh(drive)$ is within 0.2% of pre, so CC 0 to CC 1 does not jump.

Example knobs: Attack 64, Decay 80, Pitch 40, Tune 50, Noise 0, Filter 64, Dist 0, Trigger 0.

```

u(attack) = 0.503937
f_tune    = 60.408707 Hz
depth     = 5.669291 semitones
tau_p     = 0.090740 s
tau_b     = 0.136195 s
s         = 0
f_tr      = 160 Hz

```

n	f Hz	y
0	83.814360	0.000000
10	83.751440	0.055968
48	83.514083	0.040808
480	80.998689	-0.482608
2,400	72.957296	0.044642

The instantaneous frequency at $n = 2,400$ is closer to f_{tune} than the frequency at $n = 0$. That is the pitch envelope decaying. The small y values are the fold: at $n = 48$ $\sin(\text{phase})$ is about 0.5, which is 3.4 V into the cells, close to a fold root.

Same knobs, Dist 64: $drive = 4.535433$, and at $n = 48$ $y = 0.183040$.

Same knobs, Attack 127, sample $n = 10$: Trigger 0 gives $s = 0$, a sine click at $f_{tr} = 160$, $y = 0.120608$. Trigger 64 gives $s = 8$, a sine click at $f_{tr} = 1,765.184603$, $y = 0.453201$. The attack sample changed because the click rate changed.

BD2

```

trig --> sine with slow FM --> Wave, envelope toward sustain or toward 0 --> out
--> octave sine, 5 ms, level = tone ----->

```

No noise, no filter, no clip, no 16-type transient. The body can hold.

```
f_tune = 45 * (100/45) ^ u(tune)          # 45 Hz to 100 Hz
f_tr   = 2 * f_tune
tau_tr = 0.005
f_m    = 40 * (200/40) ^ u(tone)          # FM rate, 40 Hz to 200 Hz
i      = 0.12 * f_tune                     # FM depth in Hz, STAND-IN

if decay_cc >= 127:
    sustain = 0.70
    tau_b   = 0.40
else:
    sustain = 0.0
    tau_b   = decay_tau(u(decay))

env[n] = sustain + (1 - sustain) * exp(-n / (fs * tau_b))
f[n]   = f_tune * 2 ^ (bend_st * exp(-n / (fs * 0.08)) / 12) + i * sin(2 * pi * f_m * n / fs)
body[n] = fold(sin(phase[n])) * env[n]      # BD2 Wave
tr[n]   = sin(2 * pi * f_tr * n / fs) * exp(-n / (fs * tau_tr))
y[n]    = body[n] + u(tone) * tr[n]
```

fold is the Wave folder on BD2's own knob. The slow FM lets it bark: the pitch swings f_{tune} by 12% at the Tone rate, and it keeps swinging while Decay 127 holds. Tone sets both the FM rate and the octave transient level.

Pitch bend, when the step has one, uses $\tau_p = 0.08$ s STAND-IN and $\text{depth}_{\text{st}} = 12 * (2 * u(\text{bend}_{\text{cc}}) - 1)$. The Tune knob is the resting frequency. The example below has no bend.

Hold check, envelope only: Decay 127 at $n = 0$ is 1.000000, at $n = 480$ is 0.992593, at $n = 96,000$ (2 s) is 0.702021, against sustain 0.700000. Decay 0 at $n = 96,000$ is 0.000000 within $1e-5$. Decay 126 at $n = 96,000$ is 0.056280 and is not a hold. Only CC 127 holds.

Tone example, Tune 60, Tone 100, $n = 10$, envelopes of the transient only: $f_{\text{tune}} = 65.621948$ Hz, $f_{\text{tr}} = 131.243897$ Hz, $u(\text{tone}) * \text{tr} = 0.129116$. $f_m = 141.244575$ Hz, and f at $n = 10$ is 67.077722 Hz.

SD

```
trig --> pitch envelope --> shaped tone 1 --+
      --> shaped tone 2 --+--> blend --> out
      --> noise --> 4-pole low-pass, ducked --> noise envelope -->

f1      = 120 * (400/120) ^ u(tune)          # 120 Hz to 400 Hz
detune_st = -8 + 16 * u(d_tune)              # -8 to +8 semitones
f2      = f1 * 2 ^ (detune_st / 12)
depth   = 14 * u(pitch)                     # semitones
tau_p   = 0.01 + 0.12 * u(pitch)
tau_tone = decay_tau(u(tone_decay))
tau_n    = decay_tau(u(sn_decay))
tau_bend = max(tau_p, 0.08)                 # the bend's own time, 80 ms floor

p_env[n] = exp(-n / (fs * tau_p))
b_env[n] = exp(-n / (fs * tau_bend))        # 0 for every n when bend_st = 0
f1[n] = f1 * 2 ^ ((depth * p_env[n] + bend_st * b_env[n]) / 12)
f2[n] = f2 * 2 ^ ((depth * p_env[n] + bend_st * b_env[n]) / 12)
fc_n   = 1000 * (12000/1000) ^ u(tone)      # noise low-pass, 1 kHz to 12 kHz
t1[n] = shape(phase1, 2) * exp(-n / (fs * tau_tone))
t2[n] = shape(phase2, 2) * exp(-n / (fs * tau_tone))
lp[n] = ladder(duck[n] * x[n], fc_n)
nz[n] = u(snappy) * lp[n] * exp(-n / (fs * tau_n))
y[n] = (1 - u(tone)) * t1[n] + u(tone) * t2[n] + nz[n]
```

Example, noise left out (Snappy 0): Tune 70, D-Tune 90, Snappy 0, SN Decay 50, Tone 64, Tone Decay 60, Pitch 30.

```
f1      = 233.014061 Hz
f2      = 282.574688 Hz
detune_st = 3.338583 semitones
tau_tone = 0.067049 s
blend    = 0.503937
```

$\text{shape}(\text{ph}, k) = \tanh(k * \sin(\text{ph})) / \tanh(k)$, a sine bent toward square, here at a fixed $k = 2$, STAND-IN. SD has no Wave knob. The noise goes through a 4-pole low-pass, four one-poles at $g = 1 - \exp(-2 \pi f_{c_n} / f_s)$ with feedback 1.6 from the last pole, output times 2.6 so the pass band is unity. Tone sets its cutoff, so Tone 0 is dark. Resonance can pump: a peak follower on $|lp|$ (instant attack, 10 ms release) ducks the noise input, $\text{duck} = 1 - 0.25 * \min(1, \text{follow})$. It is a stand-in compressor, not a real one. Snappy is the noise level, on the short decay curve.

At $n = 20$, $t_1 = 0.899876$, $t_2 = 0.943686$, $y = 0.921953$. Both tones are in the sum. Tone CC 0 would output t_1 . Tone CC 127 would output t_2 .

Step bend on SD is a pitch drop on top of Tune with its own envelope. It is not a deeper Pitch knob, and there is no second oscillator. While bend_st is not 0 its time is the Pitch time, but never under 80 ms, so Pitch 0 with a bend still swoops. $\text{bend_st} = 0$ adds nothing, and the floor does nothing. Tune 70, Pitch 0, $\text{bend_st} = +12$: $f_1[0] = 466.028122$ Hz, and at $n = 3,840$ (80 ms) $f_1 = 300.694079$ Hz. On the Pitch time alone (10 ms at Pitch 0) it would already be 233.068249 Hz, the same as Tune within 0.06 Hz. Pitch 127 and $\text{bend_st} = +12$: tau_bend is $\text{tau_p} = 0.13$ s, and at $n = 6,240$ $f_1 = 404.878529$ Hz.

RS

```
trig --> short sine --> out

f  = 250 * (2500/250) ^ u(tune)      # 250 Hz to 2,500 Hz
tau = 0.012                          # fixed, no decay knob
y[n] = sin(2 * pi * f * n / fs) * exp(-n / (fs * tau))
```

Tune 40: $f = 516.298233$ Hz. At $n = 8$, $y = 0.507608$.

CY

```
trig --> six squares, inharmonic --+--> mix --> band-pass --> decay --> out
--> noise -----+
```

The metal is six band-limited squares (PolyBLEP) at inharmonic ratios, averaged. No sample. STAND-IN ratios, mix, band, and makeup:

```
ratios = 1, 1.4471, 1.6170, 1.9265, 2.5028, 2.6637
f0      = 180 * (900/180) ^ u(tune)      # 180 Hz to 900 Hz
mix      = 0.1 + 0.8 * u(tone)           # noise share, metal against noise
tau      = decay_tau(u(decay))

metal[n] = mean over r of square(f0 * r, n)
y[n]      = 2.0 * bandpass((1 - mix) * metal[n] + mix * x[n], 5,500 Hz, Q 0.8) * exp(-n / (fs * tau))
```

Each square starts at phase 0 on the trigger. The band-pass is the RBJ constant 0 dB peak form. Tune moves the stack, not the noise level. Tone is the mix.

Tune 64: $f_0 = 405.050673$ Hz. Tone 70: $\text{mix} = 0.540945$.

OH and HH

One colour, two decays, one choke.

```
f0      = 250 * (1000/250) ^ u(hh_tune)    # 250 Hz to 1,000 Hz, CC 73
tau_oh  = decay_tau(u(oh_decay))
tau_hh  = decay_tau(u(hh_decay))
y[n]    = 2.5 * bandpass(0.65 * metal[n] + 0.35 * x[n], 8,000 Hz, Q 1.0) * exp(-n / (fs * tau))
```

Each hat is the CY six-square metal at f_0 , mixed with noise at a fixed 0.35 share, through a band-pass at 8 kHz, times its own exp decay. STAND-IN mix, band, and makeup. Tune moves the stack, not the noise level. No hat filter knob.

A closed-hat trigger sets the open-hat envelope to 0 on that sample, and the open hat stays at 0 until the next open trigger. It does not shorten the closed hat.

Example colour, HH Tune 60: $f_0 = 481.257820$ Hz. OH Decay 100: $\text{tau_oh} = 0.276649$ s. HH Decay 40: $\text{tau_hh} = 0.033008$ s. An open hat that has run 200 samples has envelope 0.985052. On the closed trigger that envelope is 0.

CL

```
f = 400 * (3000/400) ^ u(tune)          # 400 Hz to 3,000 Hz
tau = decay_tau(u(decay))
y[n] = sin(2 * pi * f * n / fs) * exp(-n / (fs * tau))
```

Tune 50, Decay 30: f = 884.244363 Hz, tau = 0.023160 s. At n = 12, y = 0.972955.

CP

```
trig --> noise --> high-pass --> N bursts, 3 ms, 10 to 12 ms apart, panned --> stereo
      --> one-pole low-pass, delayed tail -----> both channels

s      = min(15, floor(trigger_cc * 16 / 128))    # CC 77, ASSUMED type
count  = 1 + floor(data_cc * 8 / 128)             # 1 to 8, ASSUMED, no CC
f_hp   = 300 * (1.2 ^ s)                          # 300 Hz to 4,622 Hz
gaps   = 480, 576, 528, 504, 552, 576, 480       # samples, 10 ms to 12 ms
tau_tr = 0.003                                    # fixed, Decay never stretches a burst
fc     = 400 * (6000/400) ^ u(filter)
tau    = decay_tau(u(decay))

hp[n]   = x[n] - lowpass(x[n], f_hp)
burst_i[n] = hp[n] * exp(-n_b / (fs * tau_tr))    # n_b = n - start_i, 0 before the start
tail[n]  = lowpass(hp[n], fc) * exp(-(n - t_0) / (fs * tau)) for n >= t_0, else 0
t_0     = start_(count-1) + 528
```

Burst i starts at the sum of the first i gaps: 0, 480, 1,056, 1,584, 2,088, 2,640, 3,216, 3,696. Each burst is the same high-passed noise under its own fixed 3 ms decay. Pan of burst i, count > 1: p = -1 + 2 * i / (count - 1). One burst uses p = 0. The tail is one filtered delay: the same high-passed noise through the one-pole at fc, opening one gap (11 ms) after the last burst starts and decaying on the drum law, times 0.707107 on each channel. The one-pole runs from the hit, so it is settled when the tail opens. STAND-IN.

```
yL[n] = u(attack) * sum_i burst_i[n] * gL(p_i) + tail[n] * 0.707107
yR[n] = u(attack) * sum_i burst_i[n] * gR(p_i) + tail[n] * 0.707107
```

Data CC 48 gives count 4. At any one sample, burst i over burst i - 1 is exp(gap / 144): the bursts share one noise and one 3 ms curve.

LTC, MTC, HTC

Each voice:

```
trig --> sine with slow FM --> Wave, ring --> out
      --> noise, if that voice's switch is on, shared level
```

Tom mode is the sine through that tom's own Wave folder, with the BD2 FM term at a quieter depth and a fixed rate: f[n] = f_tune * bend + 0.04 * f_tune * sin(2 * pi * 50 * n / fs). STAND-IN. Conga mode adds a second sine at 2.30 * f_tune, gain 0.35. ASSUMED. The manuals say the switch changes tom and conga and do not describe the waveform.

```
LTC f = 70 * (180/70) ^ u(tune)
MTC f = 100 * (280/100) ^ u(tune)
HTC f = 140 * (400/140) ^ u(tune)
```

Below CC 127 Decay is the shared decay law. At CC 127 a tom rings and then ends. It does not hold, because a tom that never ends is a stuck voice. Only BD2 drones. STAND-IN:

```
if decay_cc >= 127:
    sustain = 0.55
    tau_b   = 0.35
    ring[n] = sustain + (1 - sustain) * exp(-min(n, 192000) / (fs * tau_b))
    env[n]  = ring[n] for n <= 192,000 (4 s)
    env[n]  = ring[192000] * exp(-(n - 192000) / (fs * 0.05)) after that
else:
    env[n] = exp(-n / (fs * decay_tau(u(decay))))
```

There is no choke jack. The next hit replaces the ring, as for every drum voice.

Noise, when the voice switch is on (CC at or above 64): $u(\text{tom_noise}) * x[n] * \exp(-n / (fs * 0.12))$. The 0.12 s noise decay is STAND-IN and shared. tom_noise is CC 84.

LTC Tune 40: $f_{\text{tune}} = 94.251191$ Hz ($f[0]$, before the FM moves). Decay 127 at $n = 96,000$: envelope 0.551484, sustain 0.550000. At $n = 192,000$ (4 s) it is 0.550005. 50 ms into the release, at $n = 194,400$, it is 0.202335. It is under $1e-3$ at $n = 207,144$ (4.32 s), and the tom ends there. TOM_NOISE 33: level 0.259843. On the first noise draw that is -0.136958, and on the TO/CO left channel at $p = -0.7$ it is -0.133176.

Bend uses the BD2 bend paragraph.

CB

```
trig --> square at f ----+
      --> square at 1.015 f --> average --> decay --> out
```

The two squares are the brief. Detune ratio 1.015 is STAND-IN. A square is the odd harmonics $k = 1, 3, 5, \dots$ while $k * f$ is under Nyquist and $k \leq 15$, amplitude $1/k$, divided by the sum of those amplitudes.

```
f = 300 * (1200/300) ^ u(tune)          # 300 Hz to 1,200 Hz
tau = decay_tau(u(decay))
y[n] = 0.5 * (square(f) + square(1.015 * f)) * exp(-n / (fs * tau))
```

Tune 48, Decay 70: $f = 506.607352$ Hz, second square 514.206462 Hz, $\tau = 0.095560$ s. At $n = 15$, $y = 0.404500$.

Sixteen panel steps, if a UI wants them, are Tune CC values $\text{round}(i * 127 / 15)$ for i from 0 to 15. The voice reads the continuous CC.

MA

```
trig --> noise --> one-pole lowpass at 1,500 Hz --> decay --> main
```

No tune knob. Colour is fixed. STAND-IN $f_c = 1,500$ Hz. The filter is the lowpass in the shared section, held at that cutoff. Decay:

```
tau = decay_tau(u(decay))
y[n] = lowpass(x[n]) * exp(-n / (fs * tau))
```

Decay 55: $\tau = 0.056163$ s. First noise draw $x = -0.527089$. $a = \exp(-2 * \pi * 1500 / fs) = 0.821725$. With filter state 0, $y[0] = -0.093967$ before the envelope, and the envelope at $n = 0$ is 1. On the main, at the equal-power centre, each side is -0.066445.

Lead and bass

```
note --> band-limited saw --> one-pole lowpass --> envelope --> main
```

Waveform ASSUMED: saw, harmonics $k = 1$ to 8, amplitude $1/k$, skipped when $k * f$ is at or above Nyquist, then divided by the sum of the weights used.

Pitch is MIDI note 36 to 72, which is the appendix range for CV1 and CV23. $\text{Hz} = 440 * 2^{((\text{note} - 69) / 12)}$. Note 60 is 261.625565 Hz. Note 36 is 65.406391 Hz. Note 72 is 523.251131 Hz.

The panel entry in both manuals is step buttons 1 to 13 for pitch class C through the next C, and buttons 14 to 16 for the octave. ASSUMED map onto notes 36 to 72: octave 0, 1, 2 from buttons 14, 15, 16, pitch class 0 to 12 from buttons 1 to 13, $\text{note} = 36 + 12 * \text{octave} + \text{class}$, then clamp to 72. A class of 12 is the upper C, so some combinations land on the same note. That collision is accepted.

A rest, the A/B silent step, releases the note that is sounding. It does not cut it to 0. A rest on a voice that is not sounding opens nothing. A tied note keeps the envelope open and does not retrigger.

Envelope: attack is one sample to 1 on a new note, hold while the note is tied, release $\tau = 0.03$ s STAND-IN when the note ends.

Lead tone, ASSUMED, one-pole lowpass:

$$fc = 200 * (8000/200) ^ u(lead_tone)$$

Bass tone is the manual's cutoff, and the same u is the CV3 value, scaled STAND-IN as 0 V to 5 V for a jack: $cv3 = 5 * u(bass_tone)$.

$$fc = 80 * (4000/80) ^ u(bass_tone)$$

Lead example, note 60, before the filter, n = 25 samples after a trigger, harmonics 1 to 8: saw = 0.387776. Lead tone CC 80: $fc = 2,042.686148$ Hz.

Bass accent uses g_accent on the bass voice in INT. The drums' three-level accent is the same table.

Step bend and the Pitch knob

BD1's Pitch knob is both time and depth, one u, as the instrument chapter states. SD's pitch control is depth, with the time law above. Step bend is an extra signed depth on BD1, BD2, SD, LTC, MTC, and HTC only. On SD it has its own envelope with the 80 ms floor, written in the SD equation. CY, RS, hats, clave, clap, cowbell, and maracas have no step bend. EXT ignores step bend.

Init kit (INIT)

STAND-IN. A balance set by ear from the knob ranges above, with no samples. A new engine loads it. reset() still clears to empty knobs, empty steps, level 1, and master 1, which is the state every named test arms from. The plugin writes its own parameters over both on the first block.

Knob CC is round($u * 127$).

Voice	Knobs, u (CC)	Level
BD1	Tune 0.22 (28), Pitch 0.55 (70), Decay 0.28 (36), Attack 0.45 (57), Dist 0.15 (19), Noise 0.08 (10), Filter 0.35 (44)	0.85
BD2	off	0
SD	Tune 0.48 (61), Detune 0.12 (15), Pitch 0.35 (44), Tone Decay 0.22 (28), SN Decay 0.30 (38), Snappy 0.62 (79), Tone 0.45 (57)	0.75
RS	Tune 0.62 (79)	0.4
CP	Data 48 (4 bursts), Attack 0.7 (89), Decay 0.25 (32), Filter 0.55 (70)	0.7
HH	Decay 0.08 (10), Tune 0.7 (89)	0.45
OH	Decay 0.34 (43), Tune shared with HH	0.4
CY	Decay 0.55 (70)	0.35
LTC, MTC, HTC	Tune 0.30, 0.42, 0.55 (38, 53, 70), Decay 0.32, 0.28, 0.24 (41, 36, 30)	0
CL, MA, CB	default	0
Lead, bass	default, no steps	1

Every knob not listed keeps its Knobs default, including Wave at CC 32 on BD1, BD2, and the toms. Master 0.7, 120 BPM, 16ths, INT.

Pattern "INIT", length 16, every track length 16, no step on. The factory patterns are cleared for now; tests that need steps arm their own.

CHOICE: the closed hat skips the open-hat steps. A closed trigger on the same sample chokes the open hat, so a closed hat on every step would leave the open hat silent.

Not in the engine, so not in the kit: a hat filter knob, a master Drive, and a global Accent amount. The sheet called the BD1 and SD sweep "Bend"; here that is the Pitch knob. Step bend stays off. One bar peaks at 0.943890 on the main.

Cleared voice

A voice that has never been triggered in the block under test outputs 0 on every sample. A drum step that is off does not trigger. A note-track rest does not open the gate. Neither event produces a new nonzero sample from a cleared voice.

Build guide (2026-10-06)

This tree builds the voice engine, the design-pack PDF, and a placeholder VST3. `make test` stays on the engine and the clock-switch law. It does not compile JUCE.

This tree

The voice engine is `engine/`. Panel art is unpainted. The editor in `plugin/` is a placeholder plate for listening in a host. JIDAI RACK is not changed by this repository. The equations in `SCHEMATICS.md` are the voice contract. `make test` builds `build/shogun_tests`, runs the checks in `TESTPLAN.md`, and runs the clock-switch law. That law test imports GraphForge from `~/graphforge/src`.

Rebuild the pack PDF from the markdown:

```
python3 scripts/build_design_pack_pdf.py
```

The script reads `README.md`, `METHODOLOGY.md`, `SCHEMATICS.md`, `BUILD_GUIDE.md`, `TESTPLAN.md`, and `REPO_SETUP.md`, in that order, and writes `docs/SHOGUN_Design_Pack.pdf`. It needs `reportlab`. It does not fetch audio code.

Engine

Piece	Job
Clock	INT and EXT, tempo, scale, the step counter that keeps running in EXT
Clock law	<code>forge/graphs/switch_law.json</code> . The switch chooses. Printed sample rows stay in the voice tests
Pattern	14 drum tracks, 2 note tracks, lengths, shuffle, shift, mute, per-step flags
Voices	One block per voice in <code>SCHEMATICS.md</code> , cleared to 0 until triggered
Outs	The pair map, the main sum, master after the sum
Tests	The names in <code>TESTPLAN.md</code> , against the printed 48 kHz rows

Host tempo wins while the host is playing. Otherwise the internal tempo, 60 to 180 BPM, default 120, is the clock. The display shows BPM and the step. EXT leaves both of those moving.

Trig jacks are rising edges through 1.0 V. In INT they are not read. In EXT they are the only triggers. Velocity on the edge is the accent in EXT. A gate with no velocity byte uses 127.

Live knobs apply in both modes. Pattern step on, flam, accent, and step bend apply only in INT.

Placeholder plate (2026-10-06)

`make plugin` configures `plugin/` against the JUCE 8.0.4 tree already on this machine, then builds a universal (arm64 and x86_64) VST3 and a console probe. The deployment target is 11.0. Standalone is not a target on this machine: Command Line Tools do not include `ibtool`.

```
make plugin
make install-vst
```

`make install-vst` symlinks `~/Library/Audio/Plug-Ins/VST3/SHOGUN.vst3` to the Release artefact. FL Studio 2024 needs a plugin rescan after that link exists. The binary is an instrument, MIDI input is on, and the editor is fixed at 980 by 640.

The plate is flat greys and stock sliders. Each voice is one row: name, TRIG, level, and that voice's knobs. The name selects which voice the 16 step buttons edit. INT is the default and plays a beat: BD1 and bass on steps 1 and 9, snare on steps 5 and

13, hats on the eighths. EXT stays silent until TRIG or a MIDI note. MIDI notes 36 to 51 play BD1 through bass, in voice order. The clock parameter is the only writer of INT and EXT. TRIG and MIDI call the engine trigger path and leave the switch where it is.

The wrapper resamples the engine's 48 kHz stream to the host rate. Trig jacks stay unwired, so a note cannot act as a cable that forces EXT. Saved state is the parameter tree plus 16 step masks.

`make plugin` runs `ShogunProbe`, then ad-hoc signs the bundle so the module info stays inside the signature. The probe checks INT sound, EXT silence, MIDI and TRIG while EXT stays EXT, a bass note and its release, a 44.1 kHz host rate, a quiet tail after the steps are cleared, a restored step mask and clock, and that the plate paints. It writes `/tmp/shogun-plate.png` when given that path.

Panel

The jobs may match the instrument chapter: the same knob names, the same 14 drums, the two note tracks, the INT/EXT switch, and a Trig jack on each voice. The face must not be a trace of the hardware panel. No MFB logo. No dancing-bear mark. Draw a new plate. Level and master stay off the pattern, as in both manuals.

The INT/EXT switch is a SHOGUN control. Putting it on the plate does not make it a copy of a Tanzbär control, because that machine does not have this switch.

Outputs

Pairs, post level and post accent, before master:

- BD: left BD1, right BD2
- SD/RS: left snare, right rim
- HH/CY: left open plus closed, right cymbal
- CP: stereo clap
- TO/CO: LTC, MTC, and HTC at pans -0.7, 0, and 0.7
- CB/CL: left clave, right cowbell

Main is the sum of those contributions, plus maracas, lead, and bass at center pan, then the master. A patched pair does not remove its voice from the main.

Limits

The voice is the equation set in `SCHEMATICS.md`. A STAND-IN number stays marked until a measurement replaces it. A sampled kit is outside this design. A transistor netlist is outside this design: the manuals did not provide one. JIDAI RACK does not compile this repository, and this guide does not add it.

Test plan (2026-10-06)

These tests are the contract for the engine in `engine/.tests/voices.cpp` compiles them into `build/shogun_tests`. Each one is specified against `SCHEMATICS.md` at 48,000 Hz, double precision, absolute tolerance 1e-5 on the printed decimals unless a row says otherwise.

The clock switch is also pinned in `forge/graphs/switchLaw.json`. A Trig cable cannot force EXT. INT ignores Trig jacks. EXT ignores the pattern. The numbers in this file stay the named checks. A later fit may change a printed row when `SCHEMATICS.md` and `tests/voices.cpp` change together. `make test` runs those checks and `forge/tests/test_switchLaw.py`.

A cleared voice is one that has not been triggered. Its output block is 0.

Shared pattern for the bypass tests: 16th scale, 120 BPM, so a step is 6,000 samples. Pattern length 4. Track length 4. One drum voice, BD1, with the BD1 example knobs (Attack 64, Decay 80, Pitch 40, Tune 50, Noise 0, Filter 64, Dist 0, Trigger 0). Step 0 is on. Steps 1, 2, and 3 are off. No flam. Accent index 2. No bend.

testKickBendDecays

Render BD1 from the example knobs, bend_st 0, from a trigger at $n = 0$.

The frequency at $n = 0$ is 83.814360 Hz. The frequency at $n = 2,400$ is 72.957296 Hz. f_tune is 60.408707 Hz. The later frequency is closer to f_tune than the first. y at $n = 48$ is 0.040808. y at $n = 480$ is -0.482608.

Fail if the later frequency is farther from f_tune than the frequency at the trigger, or if either printed y misses by more than $1e-5$.

testBd1SoundChangesAttack

Same knobs, Attack raised to 127. Render two triggers.

Trigger CC 0: at $n = 10$, $y = 0.120608$, click rate 160 Hz.

Trigger CC 64: at $n = 10$, $y = 0.453201$, click rate 1,765.184603 Hz.

The two y values differ. The body tune is the same in both renders, so the difference is the attack transient.

testBd2CanHold

BD2 envelope, no dependence on phase.

Decay CC 127 at $n = 96,000$ is 0.702021, and the sustain it is approaching is 0.700000. The absolute difference from 0.70 is under 0.01.

Decay CC 0 at $n = 96,000$ is 0 within $1e-5$.

Decay CC 126 at $n = 96,000$ is 0.056280. That value is not a hold. The test fails if CC 126 is treated as a hold, or if CC 127 has fallen below 0.5.

Tune 60 and Tone 100, transient only, at $n = 10$: scaled transient 0.129116. This row keeps the tone path from being a silent stub.

testSnareTwoTones

SD example knobs, Snappy 0 so the noise term is 0. Tune 70, D-Tune 90, Tone 64, Tone Decay 60, Pitch 30, SN Decay 50.

f_1 is 233.014061 Hz. f_2 is 282.574688 Hz. At $n = 20$, t_1 is 0.899876, t_2 is 0.943686, and y is 0.921953.

A second render with Tone CC 0 equals t_1 at $n = 20$. A third with Tone CC 127 equals t_2 at $n = 20$. Both partials are present at Tone 64, and the blend knob selects between them.

testHatChoke

HH Tune 60, OH Decay 100, HH Decay 40. Open hat triggers at sample 0. Envelope at sample 200, before any closed trigger, is 0.985052.

A closed-hat trigger at sample 200 sets the open envelope to 0 on that sample. Samples 200 through 500 of the open hat are 0. A later open trigger may sound again. The closed hat's own tau is 0.033008 s and is not cleared by its own trigger.

testClapBurstCount

Data CC 48, so count is 4. Trigger CC 0, so the type index is 0 and f_tr is 700 Hz. Gap is 528 samples.

Four bursts start at samples 0, 480, 1,056, and 1,584. Each gap is inside 10 to 12 ms, and no burst is nonzero before its start. At sample 1,589, burst i over burst $i - 1$ is $\exp(\text{gap} / 144)$ within $1e-9$ relative: one noise, one fixed 3 ms curve. The same run at Decay 127 gives the same four burst samples, so Decay does not stretch them.

A count of 1 produces one start and no second peak 528 samples later. The voice is not a single noise hit when count is 4: four starts exist, and they are not on the same sample.

Flam is not applied. The manuals exclude clap from flam, and this test does not put a flam on it.

testExtBypassIgnoresPattern

Mode EXT. The shared pattern has step 0 on. No Trig edge in the first step.

The BD1 output over the first 6,000 samples is 0. The clock counter still advances: after 6,000 samples the display step has moved.

Then a Trig rising edge, velocity 100, at the start of the next block, with the pattern still showing an off step or an on step. The voice fires. $g_vel(100)$ is 0.819291. y at 10 samples after that edge equals the BD1 example y at $n = 10$, which is 0.055968, multiplied by 0.819291.

The pattern accent is index 2, whose g_accent is 1. The EXT render uses 0.819291, not 1. A flam on the pattern step does not add the extra hits at 180 samples.

Switch to INT on a later step without resetting the counter. The next on-step of the pattern fires, and the Trig jack no longer fires.

testIntIgnoresTrigJacks

Mode INT. Same pattern. A Trig rising edge is placed in the middle of step 1, which is off. The voice stays 0 through that step.

Step 0, with no dependence on the Trig jack, fires. With accent index 2, y at 10 samples after the step boundary is 0.055968 times 1.000000.

Pulling the Trig cable, or leaving it plugged, does not change INT. The switch does.

testRestIsSilent

From cleared voices:

- A drum step that is off, one full step at the clock above, outputs 0.
- A lead note-track rest, note slot empty, outputs 0 for that step.
- A bass note-track rest outputs 0 for that step.

A previous hit is not part of this test. The rest does not have to choke a voice that is already ringing. Hats have their own choke test.

testIndividualOutStaysInMix

One BD1 hit, level 1, master 1, accent gain 1, the example hit. The BD pair's left channel at $n = 48$ is 0.040808. The main left channel at $n = 48$ is the same 0.040808. The pair is treated as patched.

Repeat with the pair unpatched. Main left at $n = 48$ is still 0.040808.

Master 0.5 scales the main to 0.020404 and leaves the pair at 0.040808.

Maracas does not appear on any pair. A maracas trigger with the first noise draw appears on the main only. Each side of the main is -0.066445, the printed -0.093967 at the equal-power centre.

testVoiceEndsWhenQuiet

Trigger each drum voice once, BD1 with the example knobs and the others with default knobs. Each one sounds, then ends, then outputs exactly 0 on both main channels until a new trigger. The BD1 example ends where its body envelope crosses $1e-3: \text{ceil}(-\ln(1e-3) * 48,000 * 0.136195)$ samples, within one sample. With Noise 127 it ends at the same sample and stays 0, because the noise rides the body envelope. BD2 at Decay 127 is still sounding after 10 s. A BD1 that has ended plays the example $y(10) = 0.055968$ on its next trigger.

testDecayNoonIsShort

decay_tau is 8 ms at $u = 0$, 75.902 ms at $u = 0.5$, and 720.137 ms at $u = 1$. With Decay 64 on BD2, CY, OH, HH, CL, MTC, CB, and MA, on the snare's Tone Decay and SN Decay together (Snappy 127), each voice ends at $\text{ceil}(-\ln(1e-3) * 48,000 * 0.077259) = 25,617$ samples, within one sample. The clap tail (one burst) runs the same curve from where it opens, 528 samples after the hit, so it ends at 26,145. The cowbell outputs exactly 0 on that sample.

testBd2FullHolds

BD2 at Decay 127 is still sounding after 10 s, at the sustain 0.7. The next hit starts it again at 1.

testTomFullEnds

LTC Tune 40, Decay 127. At 4 s ($n = 192,000$) it is 0.550005 and still sounding. At $n = 194,400$ it is 0.202335. It ends at $n = 207,144$. A second hit at 2 s starts the envelope again at 1.

testSnareBendAtPitchZero

SD Tune 70, Pitch 0, Tone Decay 127, bend +12. f_1 is 466.028122 Hz at $n = 0$ and 300.694079 Hz at $n = 3,840$. With bend 0, f_1 stays 233.014061 Hz. With Pitch 127, at $n = 6,240$, f_1 is 404.878529 Hz.

testSoloMutesOtherVoices

BD1 (example knobs) and SD on step 0, SD soloed. For 200 samples BD1 and its pair are 0, the main is the snare alone, and the snare matches an engine without solo. The counter matches too. With solo off, the next sample of BD1 and of the main matches the engine without solo. A soloed SD on a muted track is 0 on every output, even from a direct trigger.

testInitKitIsEmptyPattern

A fresh engine, no reset(). The pattern is named INIT, length 16, every track length 16, and no step is on in all 32 steps of any track. BD1 Decay is under 0.35 ($36 / 127 = 0.283465$). BD2 and LTC are at level 0, master 0.7. One bar of INIT at 120 BPM INT is silence. Then a test beat (not a factory pattern) on the INIT kit and levels: BD1 on steps 1 and 9, SD and CP on 5 and 13, OH on 3, 7, 11, and 15, HH on the other steps of the 16. The open hat on step 3 rings above 0.5 (no choke), and the main peaks at 0.943890, under 1.

testBd1WaveAddsHarmonics

BD1 Tune 127 (140 Hz), Pitch 0, Decay 127, Attack 0, Noise 0, one hit at velocity 1. On a 2048-point Hann spectrum of the first 2,048 samples, the third harmonic over the fundamental (peak within two bins) is 0.700067 at the default Wave, over 0.03. Wave 0 and a sine with the same envelope both read 0.000026, under 0.01, so a sine-only body fails this test.

testBd2FmMovesSpectrum

BD2 Tune 127 (100 Hz), Decay 127, from 100 ms after the hit. Tone 0 against Tone 127, 2048-point spectra: the summed difference is 0.276625 of the Tone 0 sum, over 0.2. Over one second at Tone 0 the instantaneous frequency swings from 88 Hz to 112 Hz within 0.01 Hz.

testSnareFilterDarkensNoise

SD Snappy 127, SN Decay 127. Brightness is the RMS of the first difference of the ladder output over its RMS, over 2,048 samples: 0.098297 at Tone 0 and 1.294015 at Tone 127. Tone 0 is under half of Tone 127.

testHatIsInharmonic

HH Tune 0, so the lowest square is 250 Hz. Share of 2048-point spectrum energy within two bins of a multiple of 250 Hz: one PolyBLEP square 0.999330, the six-square metal 0.320088, under 0.6.

testWaveZeroIsBypass

fold(x, 0) = x bit for bit for x from -1 to 1 in steps of 0.001. BD1 Tune 127, Pitch 0, Decay 80, Attack 0, Wave 0 is $\sin(\text{phase}) * \exp(-n / (\text{fs} * \text{tau}_b))$ within 1e-9 for 4,800 samples. BD1, BD2, LTC, MTC, and HTC at Wave 0 each put under 0.01 of their 2048-point spectrum energy above 1.5 times the fundamental.

testWaveAddsHarmonics

Default Wave is CC 32. $g(1) = 0.5$ and $g(127) = 4$. On the same held hits, the share above 1.5 times the fundamental at the default Wave is 0.388 on all five body voices, over 0.05 and ten times Wave 0, and at CC 127 it is 0.033, still over twice Wave 0. A full-scale input peaks at 1 within 1e-3 after the cells at $g = 0.5, 1, 1.2, 2, 3.3, 4$, and the default.

testWaveCellMatchesSergeMiddle

C6 at $g = 1$ against serge_middle GOLDEN.md within 1e-6: -6 V is -0.574899, -1 V is 0.160778, -0.5 V is -0.136312, 0 is 0, and the positive side is the odd mirror. OUTPUT_GAIN times $|C6(4.707287 \text{ V})|$ is 1 within 1e-12. P(2) is the stored knot 4.104493885791202. The Lambert W residual is under 1e-12 from 0.01 V to 6 V and at 48 V.

testHatHasNoWave

Every Wave knob at 0 against every Wave knob at 127. HH, OH, CY, CP, and MA each sound and are bit-identical over 9,600 samples.

testDistBypassAndDrive

BD1 example knobs, $n = 48$. Dist 0 is $y = \text{pre} = 0.040808$, with the default Wave before it. Dist 1 is within 2e-3 of Dist 0. Dist 64 is 0.183040.

testCenterPanIsNotHalf

A maracas trigger puts 0.707107 times the maracas sample on each side of the main, not 0.5. At $n = 0$ that is -0.066445. The mid tom is equal on both TO/CO channels and on the main. With Attack 0 the clap is its tail alone: 0 on both channels for the 528 samples before the tail opens, then equal on both channels and on the main.

testShuffleSurvivesOddLength

BD1 track of length 3, every step on, shuffle 15, 120 BPM 16ths. The delay is period / 3 = 2,000 samples. The first six hits are at samples 0, 8,000, 12,000, 20,000, 24,000, and 32,000: every odd clock step is late, including track step 0 on the second pass.

Also locked by those tests

- EXT does not freeze the counter: testExtBypassIgnoresPattern reads the display after one silent step.
- A Trig cable does not select EXT: testIntIgnoresTrigJacks keeps the cable connected.
- Clap spacing is 11 ms: testClapBurstCount.

- BD2 hold is CC 127 only: `testBd2CanHold`.
- Only BD2 holds; a tom at CC 127 ends: `testBd2FullHolds`, `testTomFullEnds`.

Repository setup (2026-10-06)

Tree

Public repository: <https://github.com/martialsystems/shogun>

Working name: SHOGUN. A separate product name has not been set. The engine, the clock-switch pin, and the placeholder plate are on branch `feat/engine-voices`, cut from the design pack in this repository. The pin is in `forge/`. The plate is in `plugin/`.

This tree contains:

Path	Role
README.md	Front page and revision list
METHODOLOGY.md	Sources and the quoted choices
SCHEMATICS.md	Equations and 48 kHz rows
BUILD_GUIDE.md	PDF build, and the engine layout
TESTPLAN.md	Named tests
REPO_SETUP.md	This file
engine/dsp.h	Shared STAND-IN helpers
engine/shogun.h	Clock, pattern, knobs, and the process call
engine/shogun.cpp	Voices and the output sum
tests/voices.cpp	Named checks
Makefile	<code>make test</code> builds <code>build/shogun_tests</code> and runs the clock-switch law. <code>make plugin</code> builds the VST3
plugin/CMakeLists.txt	VST3 target. Universal Release build. No standalone target
plugin/Source/	Processor, placeholder plate, knob table, and probe
scripts/build_design_pack_pdf.py	Markdown to PDF
docs/SHOGUN_Design_Pack.pdf	The pack as one PDF
forge/engine_pin.json	GraphForge pin, clock switch only
forge/product_laws.py	Legal INT and legal EXT states passed to <code>require_law</code>
forge/graphs/switch_law.json	The switch law. A Trig cable cannot force EXT
forge/tests/test_switch_law.py	Allows both switch positions and blocks the cable override
LICENSE	All rights reserved, Martial Systems LLC
.gitignore	<code>build/</code> and <code>__pycache__/</code>

The voice engine is framework-free C++ under `engine/`. `process()` does not allocate. There is no panel SVG. JUCE is the wrapper in `plugin/` only.

The MFB manuals are not in the tree. The German PDF that was read for the quotations stays outside the repository. The pack cites it. It does not redistribute it.

PDF

```
python3 scripts/build_design_pack_pdf.py
```

The script fails if a source file contains an em dash, an en dash, or a heading that defines the work by absence. The PDF is a flowing render of the six markdown files, in the order README, METHODOLOGY, SCHEMATICS, BUILD_GUIDE, TESTPLAN, REPO_SETUP. A title line on each PDF page carries the date 2026-10-06. That date is the revision date. A clock time inside the PDF metadata is only the build time.

Collection link

martialsystems/jidai-collection names this repository under the SHOGUN heading on main. The rack plugin in that repository compiles BUSHIDO and RONIN. It does not fetch SHOGUN. This engine does not change the rack.

Git

Commits on this repository use author Martial Systems LLC and 25778085+martialsystems@users.noreply.github.com, the same author as the collection repository.

The license grants no right to copy the pack without permission from Martial Systems LLC. Short quotations in METHODOLOGY.md are the disputed parameter sentences, with the source named beside each one.

Clock switch

INT ignores Trig jacks. EXT ignores the pattern. The switch chooses. A plugged Trig cable does not force EXT.

That rule is forge/graphs/switch_law.json, loaded by forge/product_laws.py. forge/engine_pin.json pins GraphForge for this law. forge/tests/test_switch_law.py asks require_law to allow both switch positions with a cable plugged, and to block a cable that forces EXT, an INT state that reads Trig jacks, and an EXT state that fires from the pattern.

The printed 48 kHz rows stay in tests/voices.cpp. A later fit may change a row when SCHEMATICS.md and that test change together.

INT and EXT are specified in METHODOLOGY.md and SCHEMATICS.md. The engine checks are testExtBypassIgnoresPattern and testIntIgnoresTrigJacks.